

DATABASE SYSTEMS

Week 1,2

Instructor : Dr. Natash Ali Mian





Switch off mobile phones during lectures, or put them into silent mode

CONTENTS

- ❑ Class Introduction
- ❑ Instructor's Introduction
- ❑ Introduction to Course
- ❑ Student Guidelines

INTRODUCE YOURSELF

Name

Expectation from this Course

Expectations from Instructor

Why CS?

Any suggestions or comments regarding the course

INSTRUCTOR'S INTRO

Education

PHD-CS

Experience

24 Years

Stay at BNU

More than 12 Years

Current Position at BNU

HOD-SE, Associate Professor

Research Areas/ Interest Software Engineering, Databases and more

COURSE OBJECTIVE

- ❑ An introductory course in Database management
- ❑ Links data and processes effectively and efficiently
- ❑ You will learn how databases are used to store, manipulate, and retrieve data
- ❑ It covers various technologies used to build and manage databases and how different applications are dependent upon databases today
- ❑ The course explains how to build robust and flexible databases
- ❑ Students will learn about the essential skills to design effective databases and to retrieve data through SQL queries

LEARNING OUTCOMES

- ❑ To learn about the limitations of traditional file processing systems.
- ❑ To be able to model an application's data requirements using conceptual modeling tools like ER diagrams and design database schemas based on the conceptual model.
- ❑ To identify functional dependencies and to resolve database anomalies by normalizing database tables.
- ❑ To use Structured Query Language (SQL) for database definition and manipulation in any DBMS
- ❑ To introduce students with transaction management concepts
- ❑ To introduce students with No SQL databases
- ❑ To introduce students with latest topics and trends in databases

TEXT BOOK

- ❑ Database Systems: A Practical Approach to Design, Implementation, and Management, 6 th Edition by Thomas Connolly and Carolyn Begg, Pearson
- ❑ Fundamentals of Database Systems 7th Edition by Ramez Elmasri, Shamkant B. Naathe
- ❑ Database System Concepts, 7th Edition by Avi Silberschatz, Henry F. Korth and S. Sudarshan

KEY TOPICS

Introduction to Databases

Database Approach

Database System Concepts and Architecture

Database Languages

Functions of a DBMS

Data Modeling using ER Model

Relational Data Model and Relational Database Constraints

Structured Query Language (SQL)

SQL Data Definition and Data Types

Specifying Constraints in SQL

Basic Retrieval Queries in SQL

KEY TOPICS

Complex SQL Queries, Views and Transactions

Advanced SQL

The Enhanced ER Model

Relational Algebra

ER-to-Relational Mapping

Normalization

NOSQL Systems

Transaction Management

Transaction Support

Concurrency Control

Database Recovery

Big Data and Data Analytics

GRADING SCHEME

Class Tests / Quiz	10
Assignments	10
Term Project	15
Mid Term Test	25
Final Exam	40

ASSIGNMENTS (10%)

You can expect 3 to 5 assignments throughout the semester

No credit for late submissions

No Assignment will be dropped

Most of the assignments will be also be interim submissions of the project

CLASS TESTS/ QUIZ (10%)

You can expect 3 to 5 tests throughout the semester

No retakes

No quiz will be dropped

Quiz may be announced/un-announced.

Always be prepared for the class test any time during the course

TERM PROJECT+PRESENTATION (15%)

Finalize groups and company by 2nd week and start working on the project

You have to submit a report and present it in the class (14th / 15th Week)

Anyone absent on the presentation day will be marked ZERO and his/ her presentation will not be rescheduled

TERM PROJECT

Irrespective of the semester work marks, Failure to submit the project will result in “F” grade

Irrespective of the semester work marks and submission of project, Failure to appear for presentation will result in “F” grade

Submission in 14th / 15th week

Presentation/Viva in 15-16th week

Further details will be circulated later

MIDTERM EXAMS (25%)

1 Mid Terms

As per university's schedule

Duration : 1 Hour 15 minutes

Syllabus covered till 8th week

FINAL EXAM (30%)

As per university's schedule

Will cover the whole of the course

Duration: 2-3 hours

GUIDELINES FOR STUDENTS (1)

- Quizzes may be announced or un-announced, so be prepared. All quizzes may be unannounced
- No evaluation will be dropped
- No Extensions in deadlines will be given
- Use of Mobile Phones is not allowed in the class (even to check time etc). Anyone found using the mobile will be requested to leave the class and you will be marked ABSENT
- Use of Mobile Phones is not allowed in the class, If mobile phone rings (due to: call, sms, alarm, reminder or any other), you will be requested to leave the class and you will be marked ABSENT.

GUIDELINES FOR STUDENTS (2)

- Students are encouraged to discuss assignments but it is extremely important that everyone works on his/her own assignment. No compromise on Plagiarism
- You may use chat bots (CHATGPT) for learning and exploring concepts. Make the best use of the latest tools to learn. That's perfectly alright.
- Don't use chat bot, language software or AI engine to generate submissions, project submission. Even a single line found will result in award of ZERO
- The cases of plagiarism will be dealt ruthlessly & will be marked Zero, or you may be eliminated from course. No Compromise
- Late comers should consult their class fellows for the missing topics, they will not be revised in the class.
- No Extensions in deadlines will be given
- Be punctual, Once the attendance is marked it cannot be changed

GUIDELINES FOR STUDENTS (3)

- You should keep a track of your attendance yourself (From CMS), no flexibility in attendance will be given
- Remember minimum attendance required to appear in final exam is 75%, **However, you are expected to attend all classes.** Don't request any flexibility or you have to face embarrassment
- You may consult me anytime for any help, discussion or guidance regarding the course. Consultancy hours will also be notified. Same will be posted outside my office
- Further guidelines will be given time to time



DATA???



WHY WE NEED A DATABASE???

- Large amounts of data (Giga's, Tera's)
 - *Can you handle it without a proper system?*
 - *Do you make folders in your computer?*
 - *Do you have folders in your cell phones?*

Why???

WHY WE NEED A DATABASE???

- Valuable data
- Performance requirements
- Multiple users accessing data
- Limited access to data (only right person should access the data)

???

Have you ever used
a Database
System???

DATABASE



DATABASE



Database is simply collection of Data

DATABASE



Precisely,
It's a
collection of
related data

BENEFITS OF USING A DB



BENEFITS OF USING DATABASE SYSTEM

- Large amounts of data can be processed
- Valuable data can be stored safely
- Data can be retrieved Efficiently
- Multiple users can access data at same time
- only the person authorized can access the data
- Multiple views of data for different users
- Redundancy Control

APPLICATIONS OF DATABASES (A FEW)

Banking

Airlines

Universities

Telecommunication

Finance

Sales

Manufacturing

Human resources

Scientific Studies

We are surrounded by Data and Database Systems

CONCLUSION

We all are maintaining a large amount of data

We all are maintaining a database (structured/ unstructured)

We all are using a database system (software)

Software engineers develop databases and database systems

Need for advance tools is increasing day by day

Databases help us in processing a lot information conveniently

Reports generated help us in decision making

Searching a specific record becomes easy and fast

Database Systems are everywhere and are extremely useful



INFORMATION ENGINEERING

INFORMATION SYSTEM ARCHITECTURE

Information System Architecture

Information Engineering

- Planning
- Analysis
- Design
- Implementation

INFORMATION SYSTEMS ARCHITECTURE (ISA)

Overall blueprint for organization's information systems

Consists of:

- Data (Enterprise Data Model – simplified ER Diagram)
- Processes – data flow diagrams, process decomposition, etc.
- Data Network – topology diagram
- People – people management using project management tools (Gantt charts, etc.)
- Events and Points in Time (when processes are performed)
- Reasons for events and rules (e.g. decision tables)

INFORMATION ENGINEERING

A data-oriented methodology to create and maintain information systems

Top-down planning approach.

Four steps: (What is SDLC?)

- **Planning**
 - Results in an **Information Systems Architecture**
- **Analysis**
 - Conceptual Data model, Detail Entity Relationship diagram
- **Design**
 - Logical database Design & Physical Database Design
- **Implementation**
 - Implementation of the relational database in any DBMS

APPROACHES TO DATABASE AND IS DEVELOPMENT

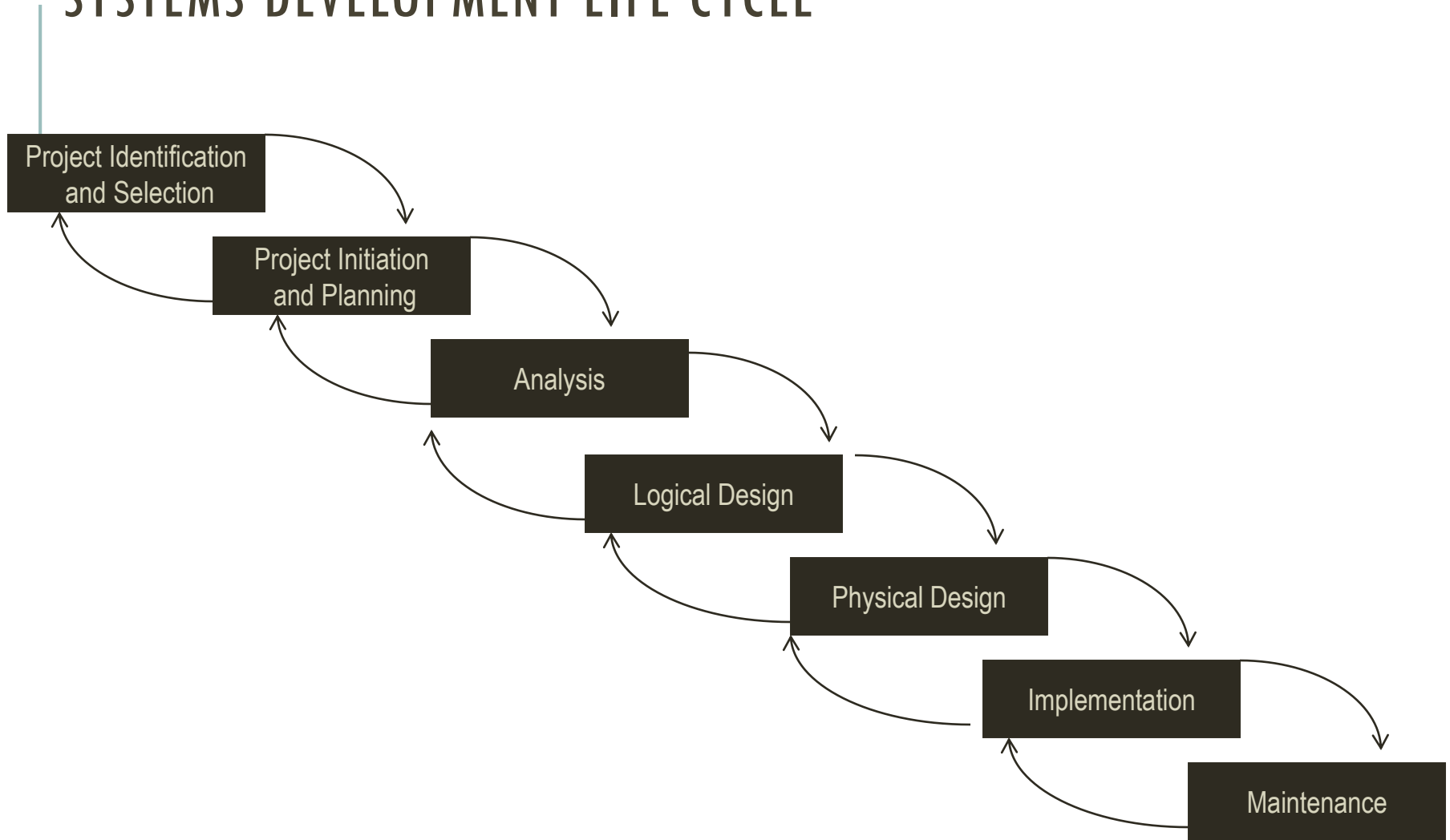
SDLC

- Systems Development Life cycle
- Detailed, well-planned development process
- Time-consuming, but comprehensive
- Long development cycle

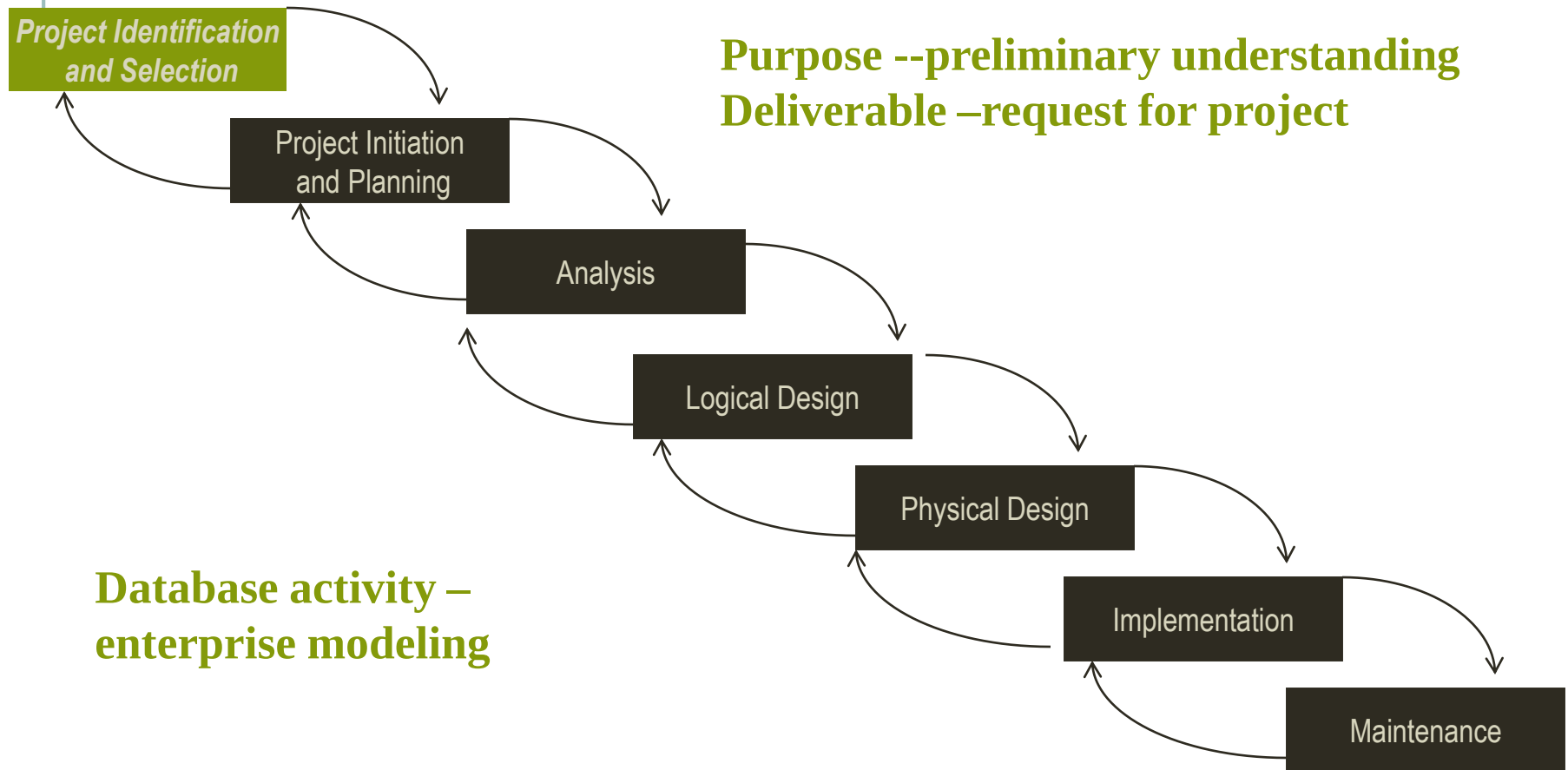
Prototyping

- rapid attempt to develop a conceptual data model
- Define database during development of initial prototype
- Repeat implementation and maintenance activities with new prototype versions

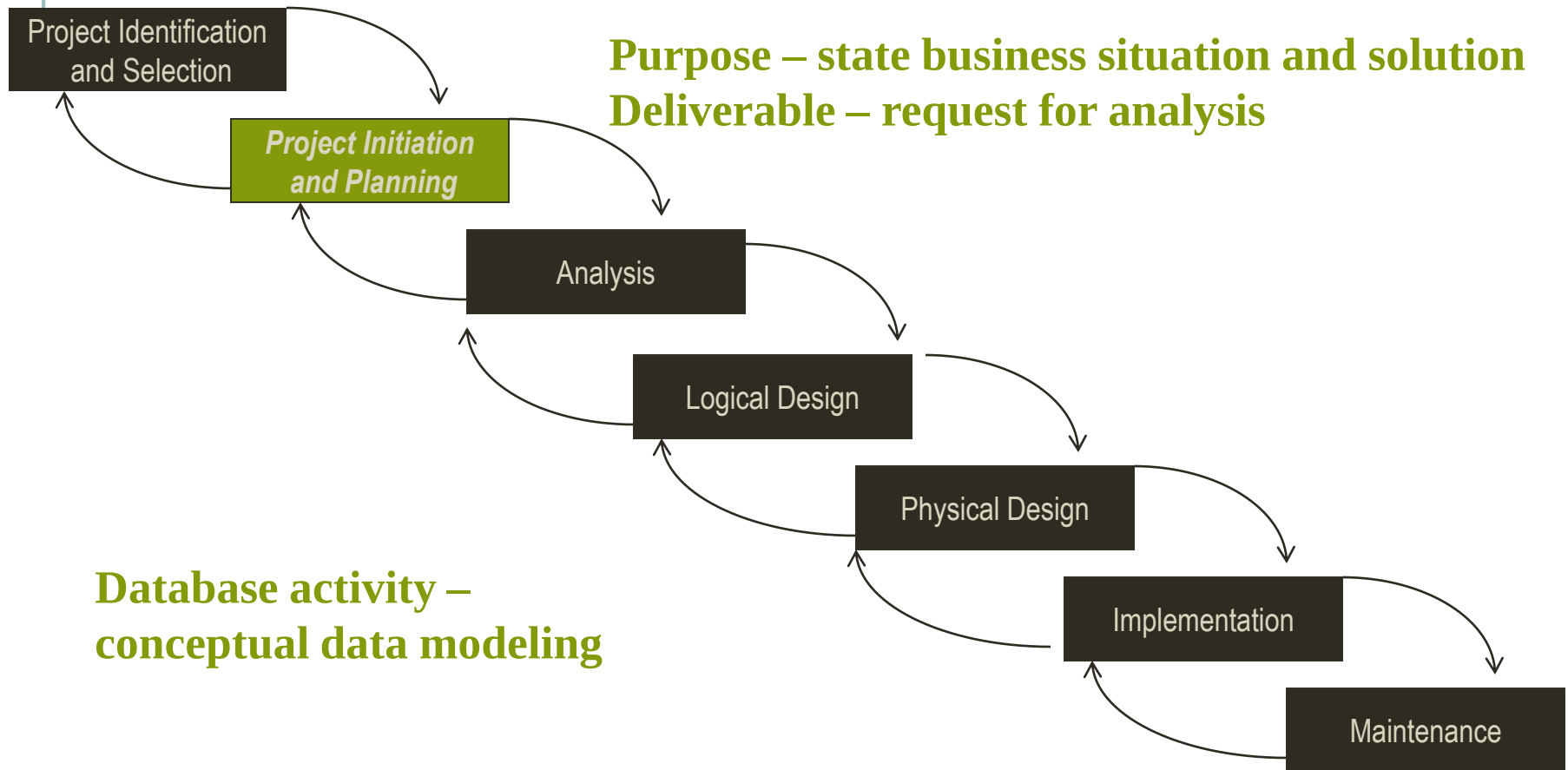
SYSTEMS DEVELOPMENT LIFE CYCLE



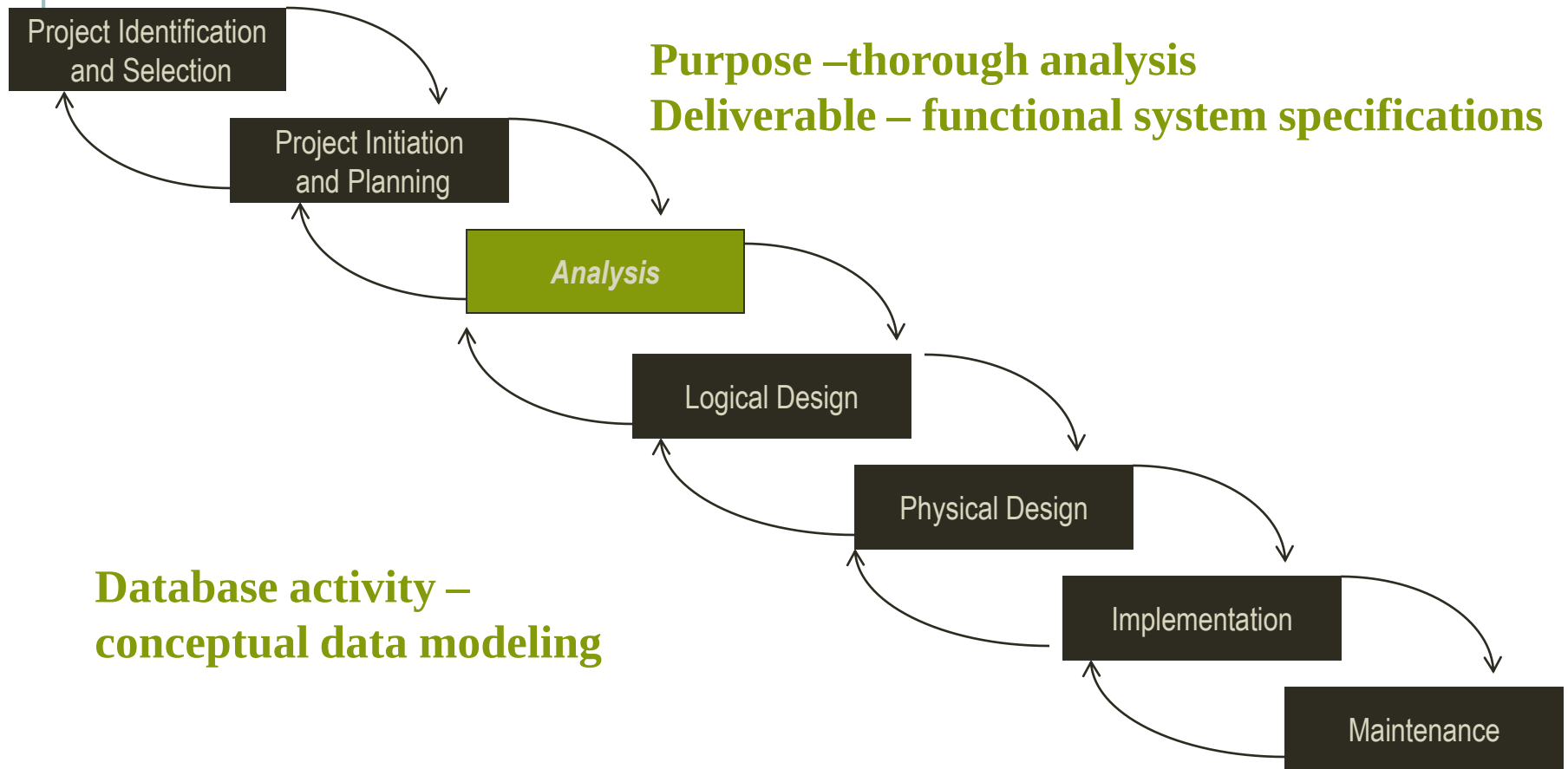
SYSTEMS DEVELOPMENT LIFE CYCLE



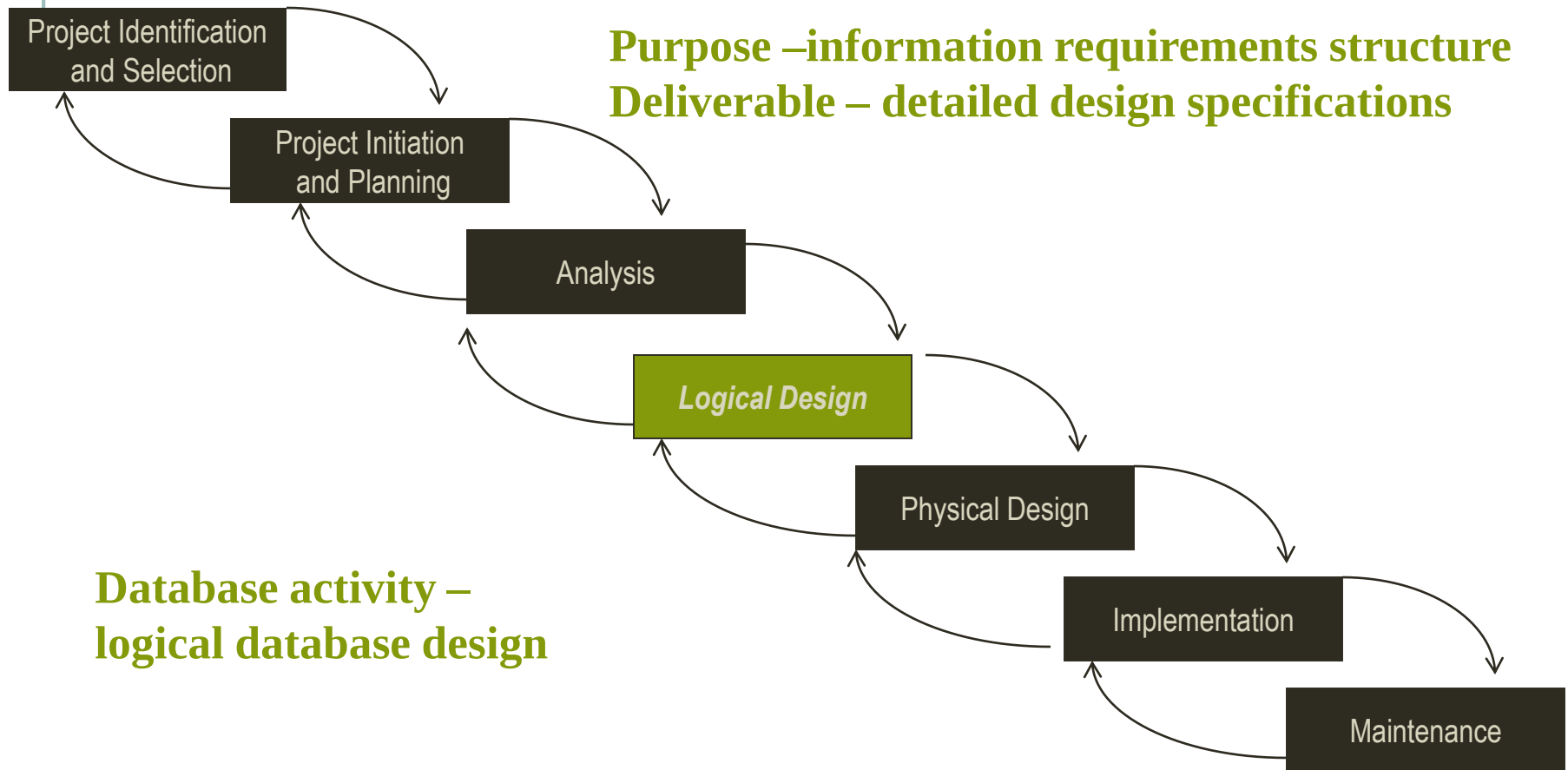
SYSTEMS DEVELOPMENT LIFE CYCLE



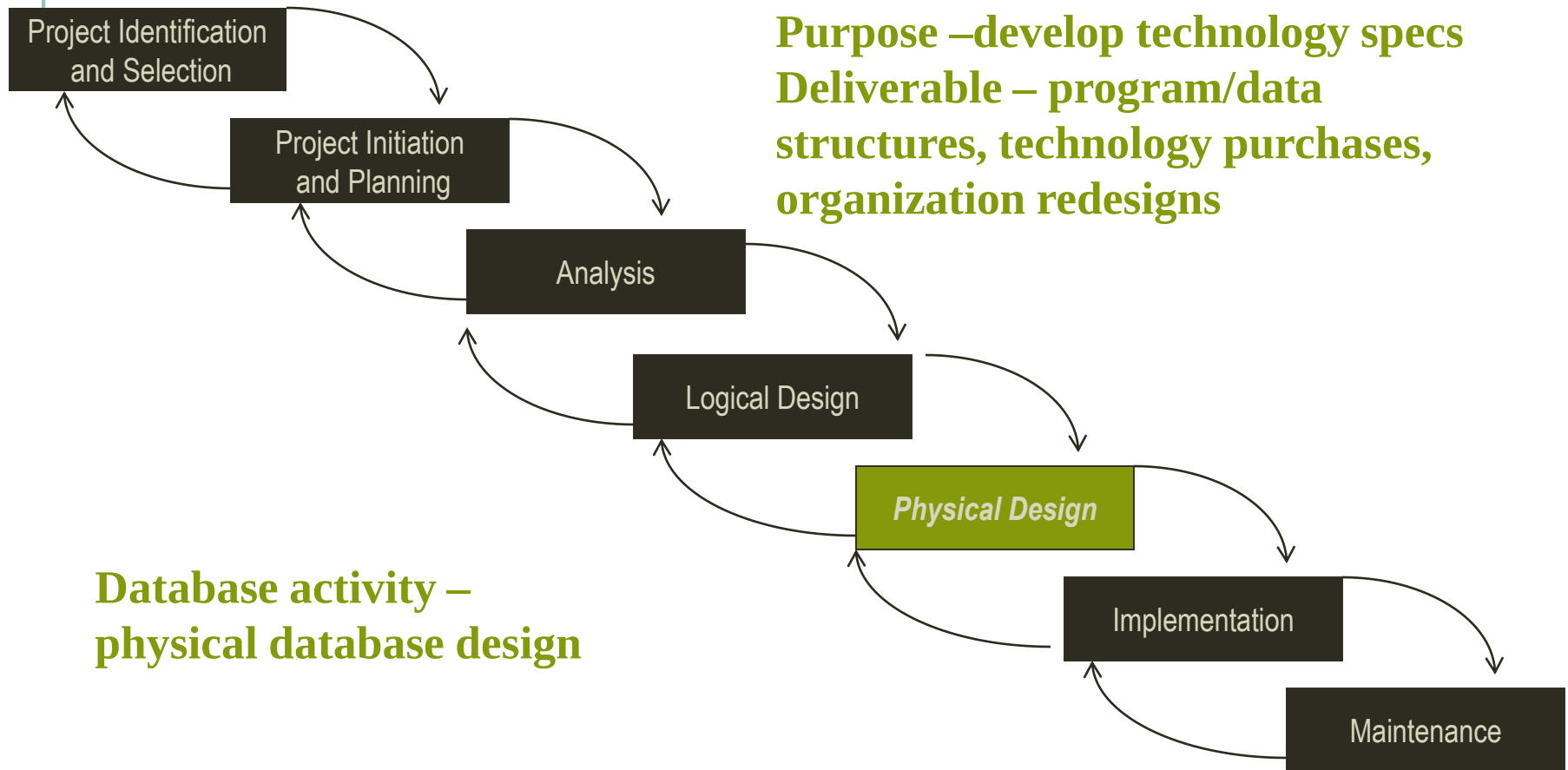
SYSTEMS DEVELOPMENT LIFE CYCLE



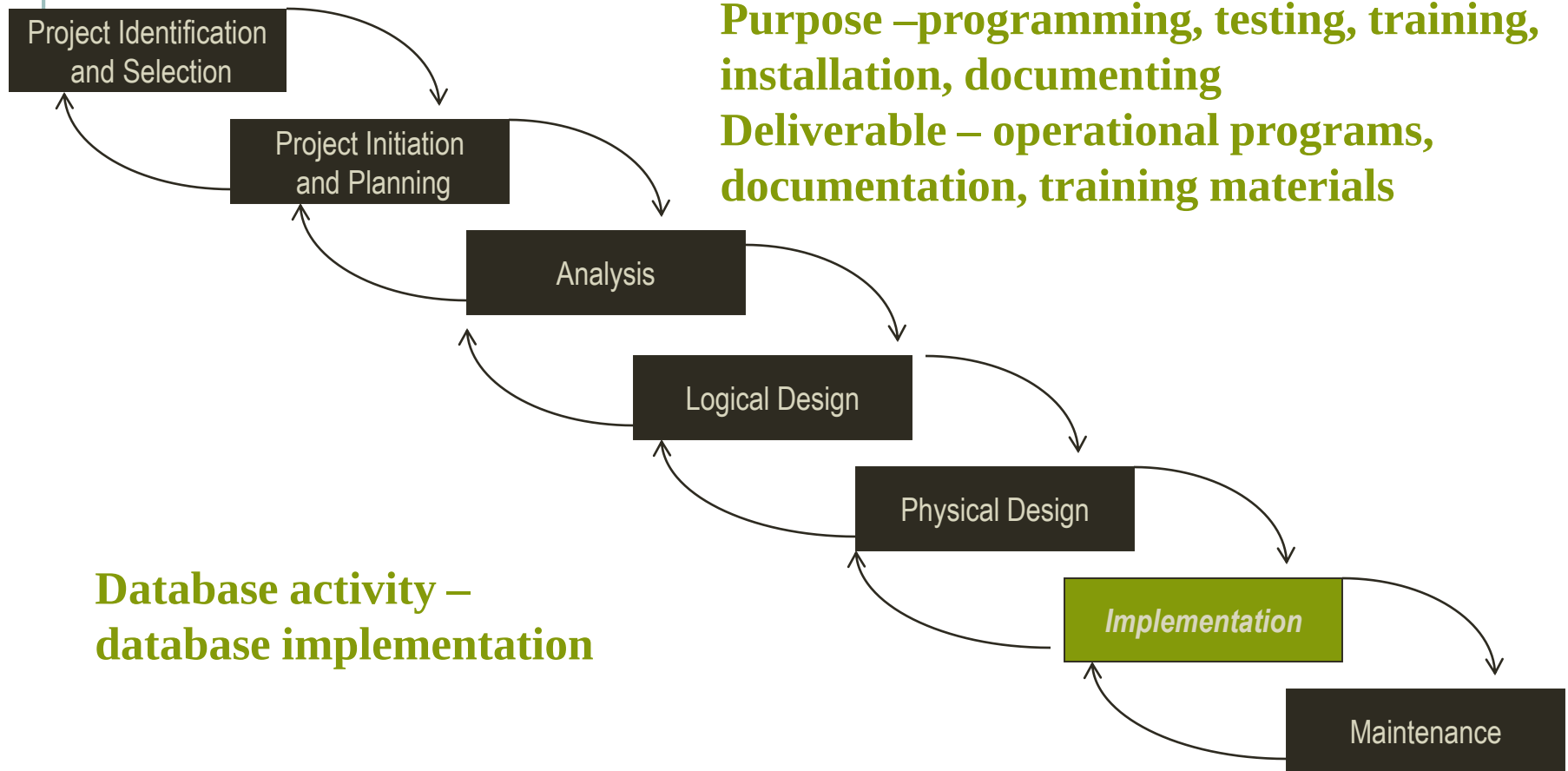
SYSTEMS DEVELOPMENT LIFE CYCLE



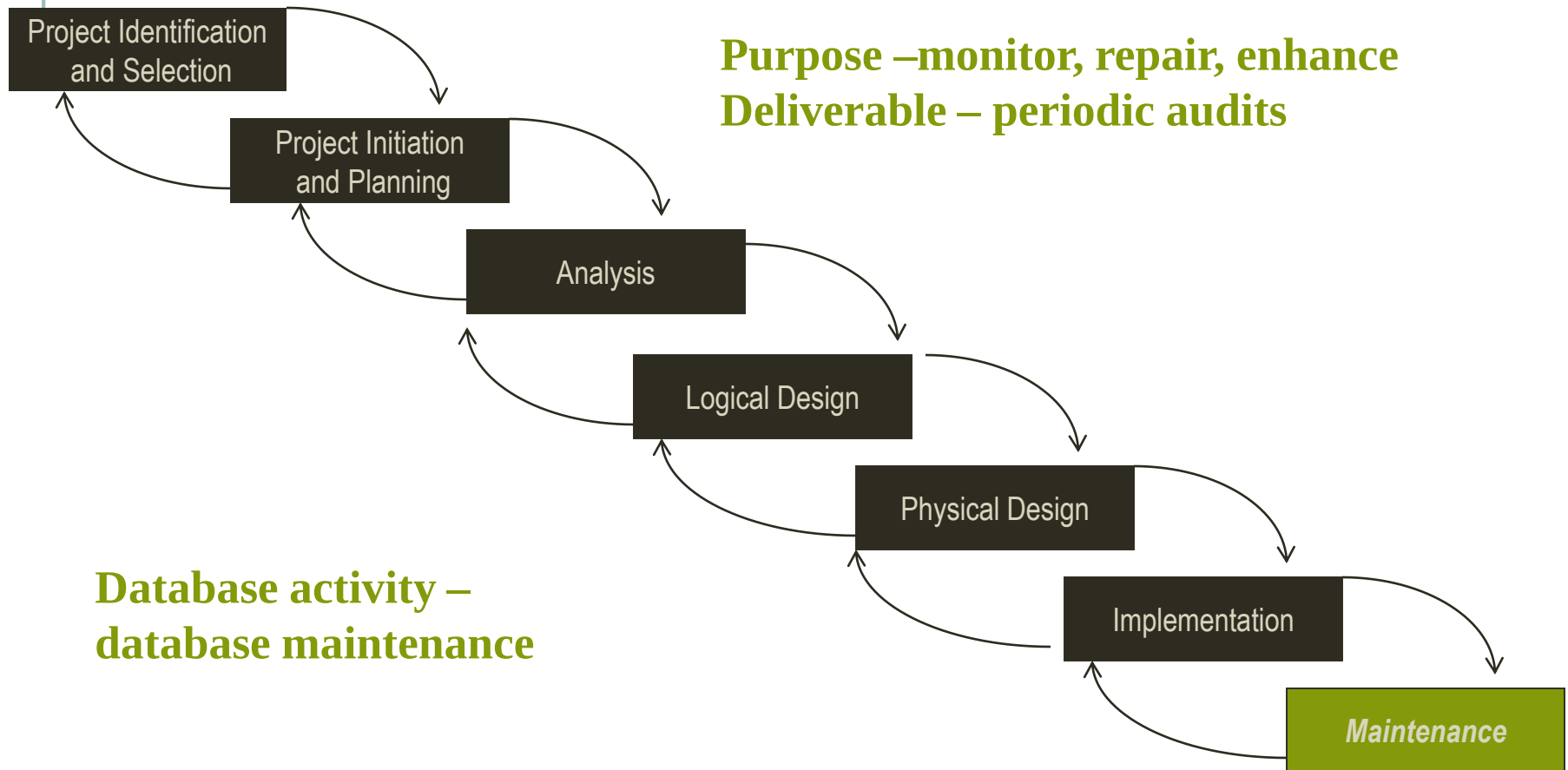
SYSTEMS DEVELOPMENT LIFE CYCLE



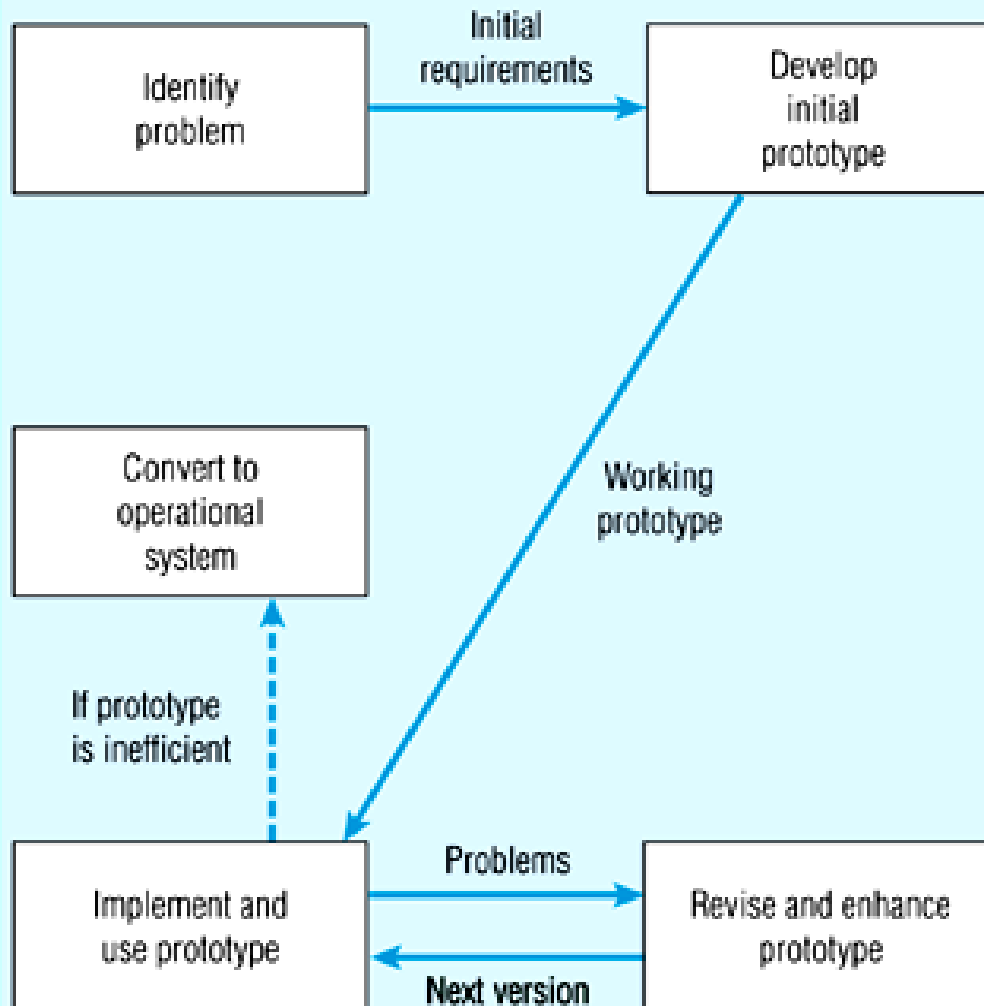
SYSTEMS DEVELOPMENT LIFE CYCLE



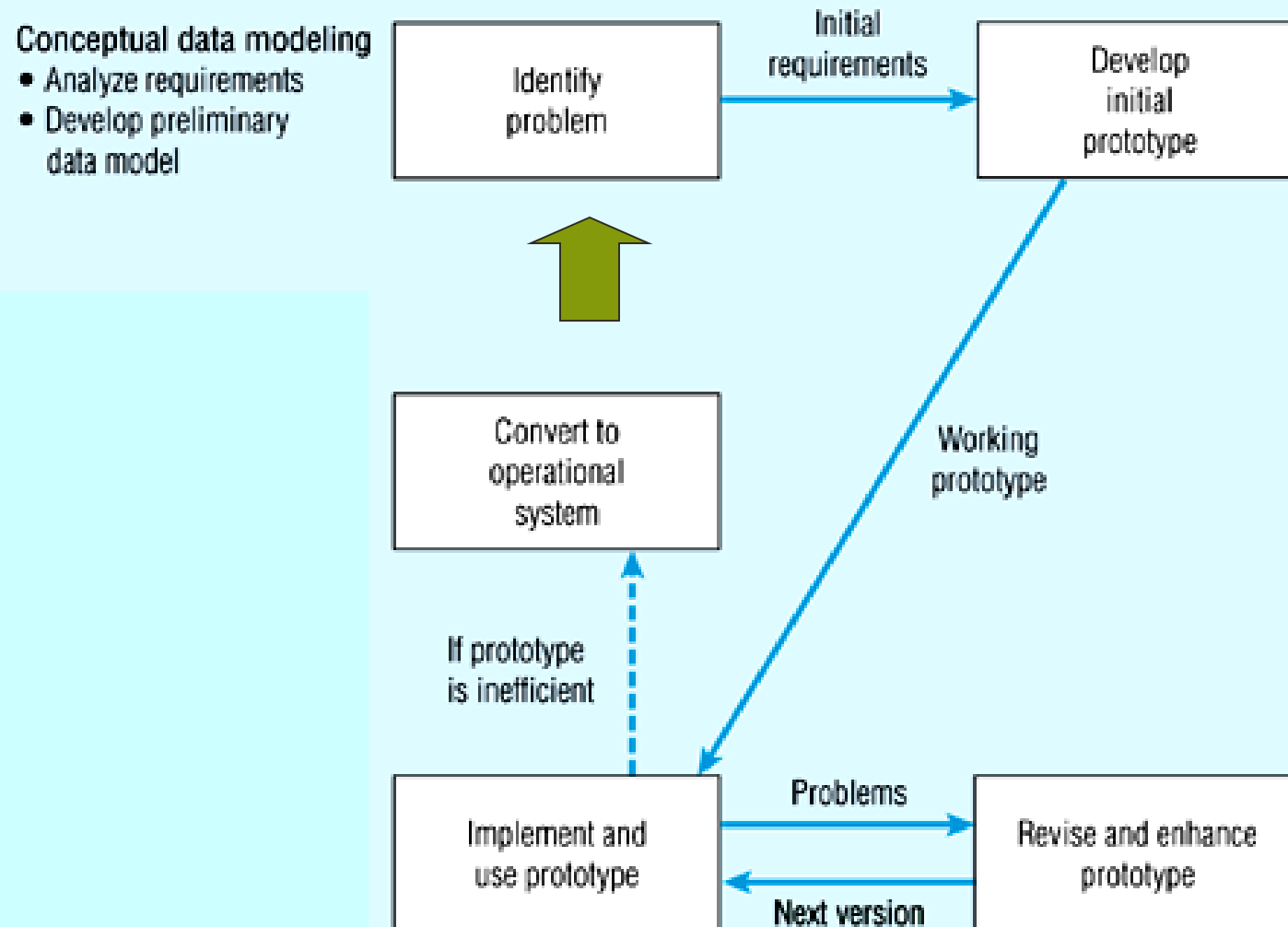
SYSTEMS DEVELOPMENT LIFE CYCLE



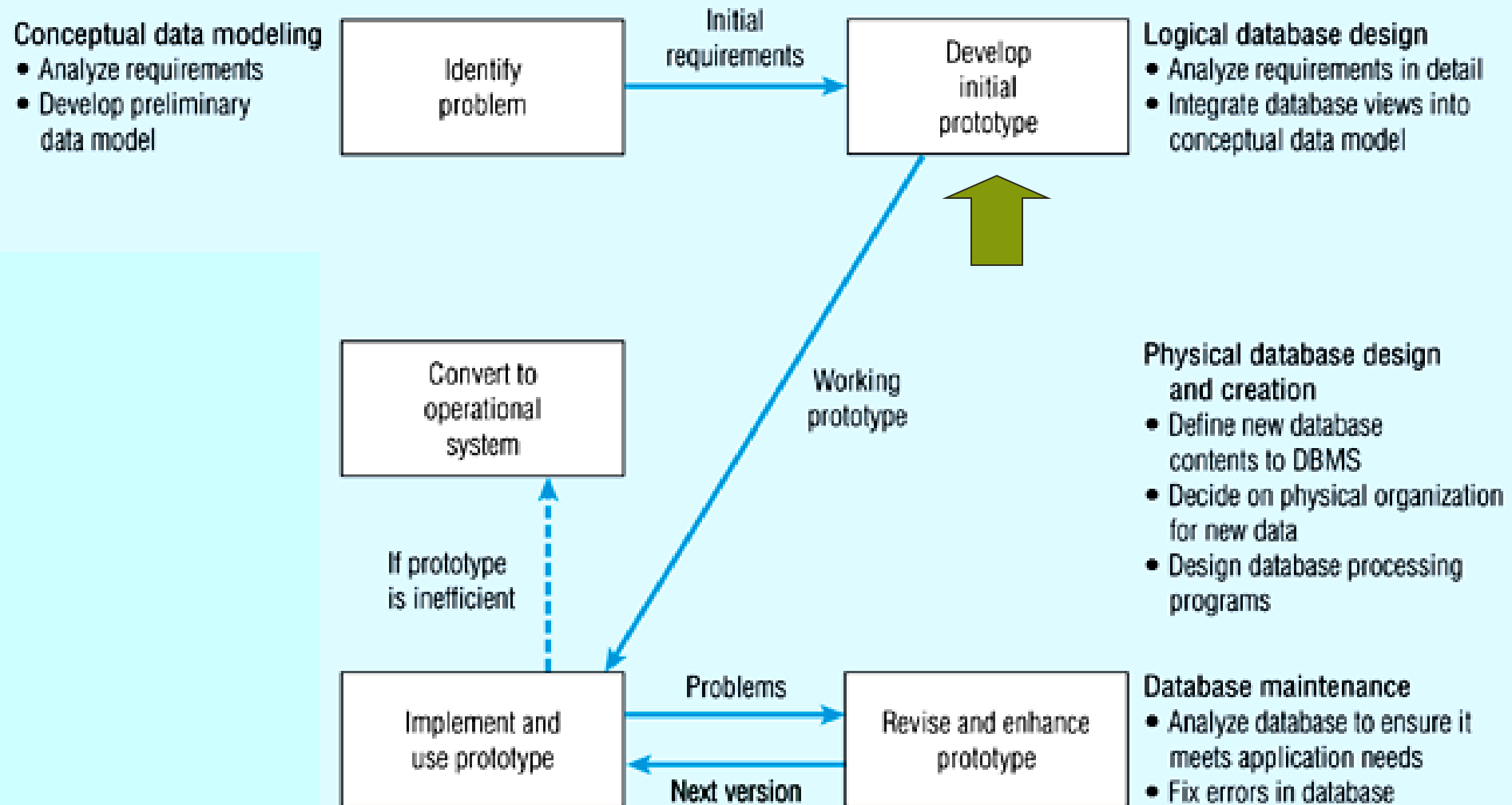
The *prototyping methodology* and Database development process



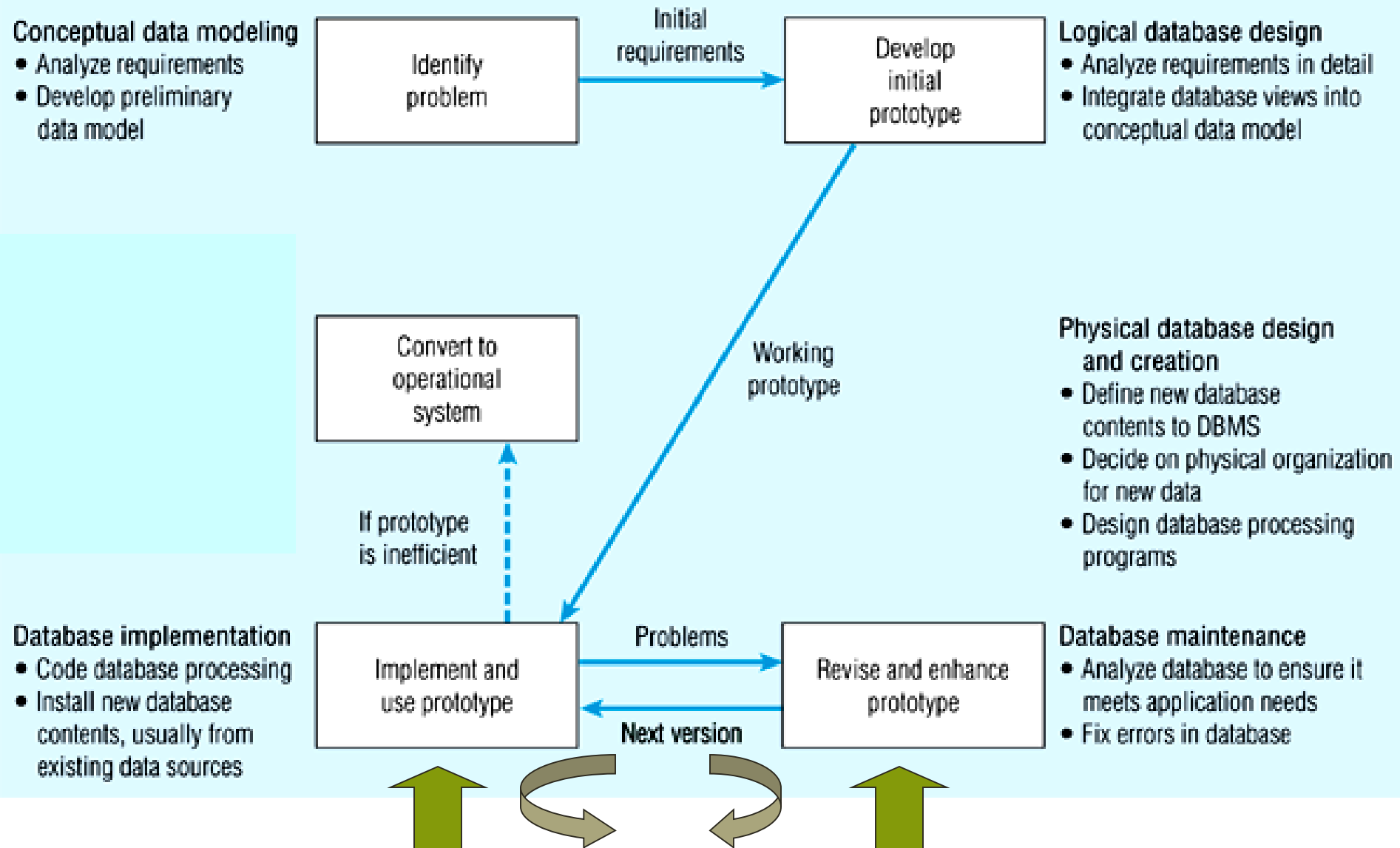
The *prototyping methodology* and Database development process



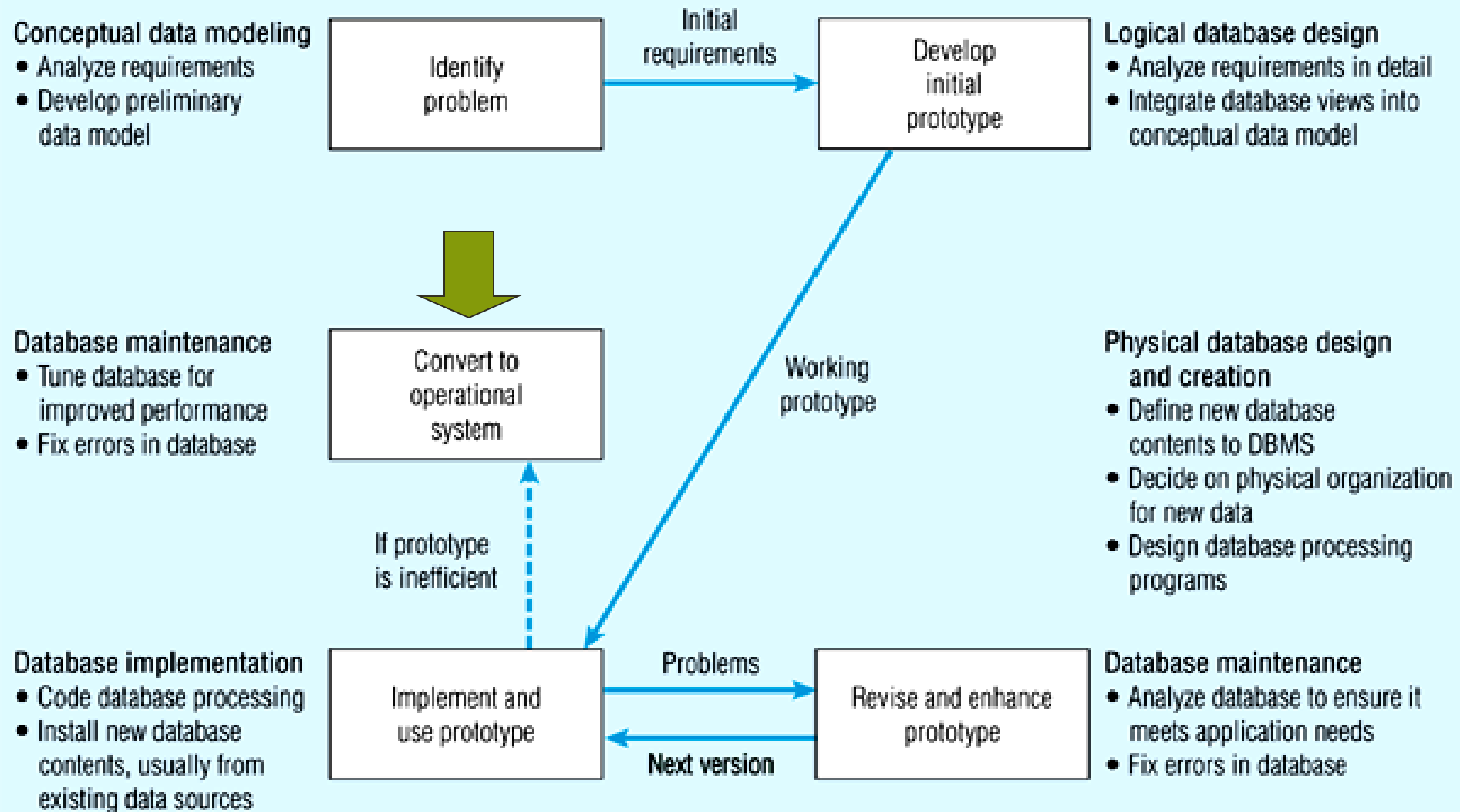
The *prototyping methodology* and Database development process



The *prototyping methodology* and Database development process



The *prototyping methodology* and Database development process



MANAGING PROJECTS: PEOPLE INVOLVED

Systems analysts

Database analysts

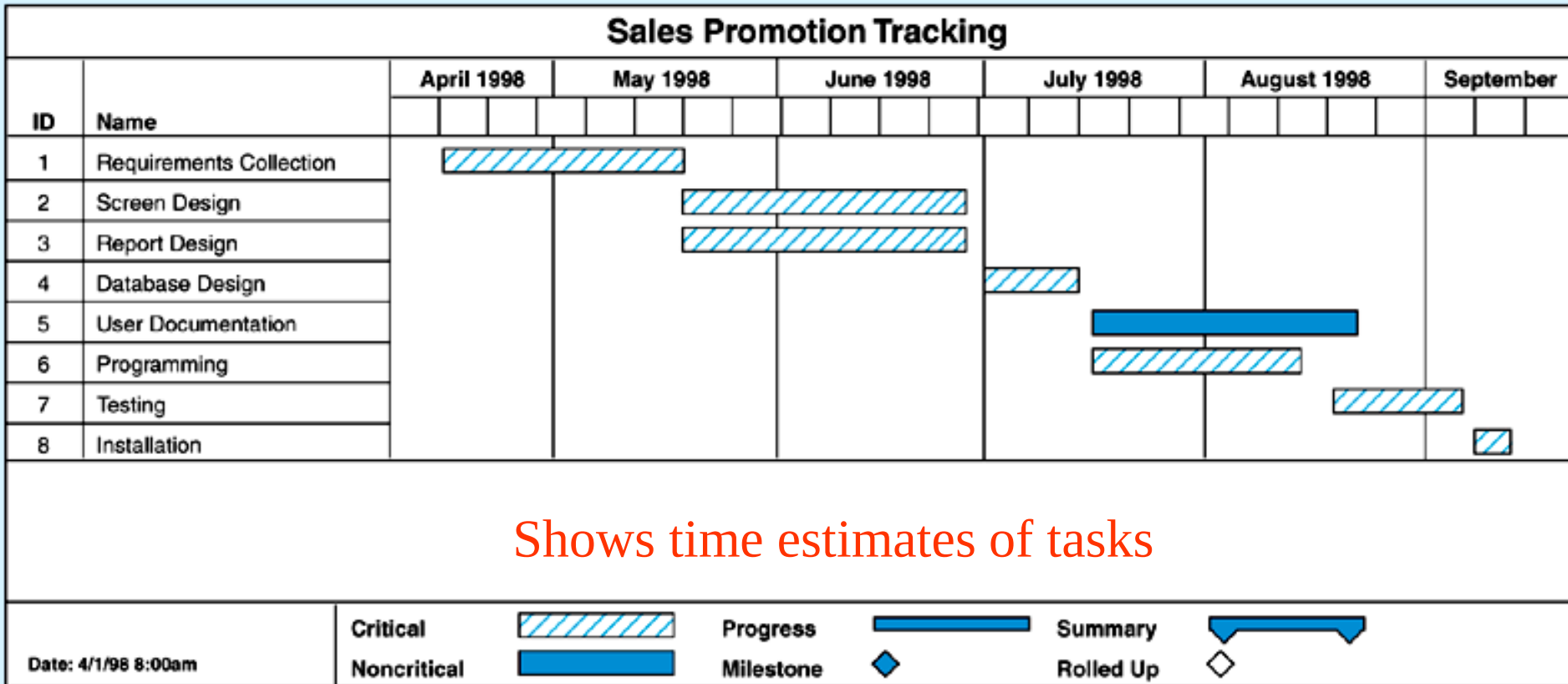
Users

Programmers

Database/data administrators

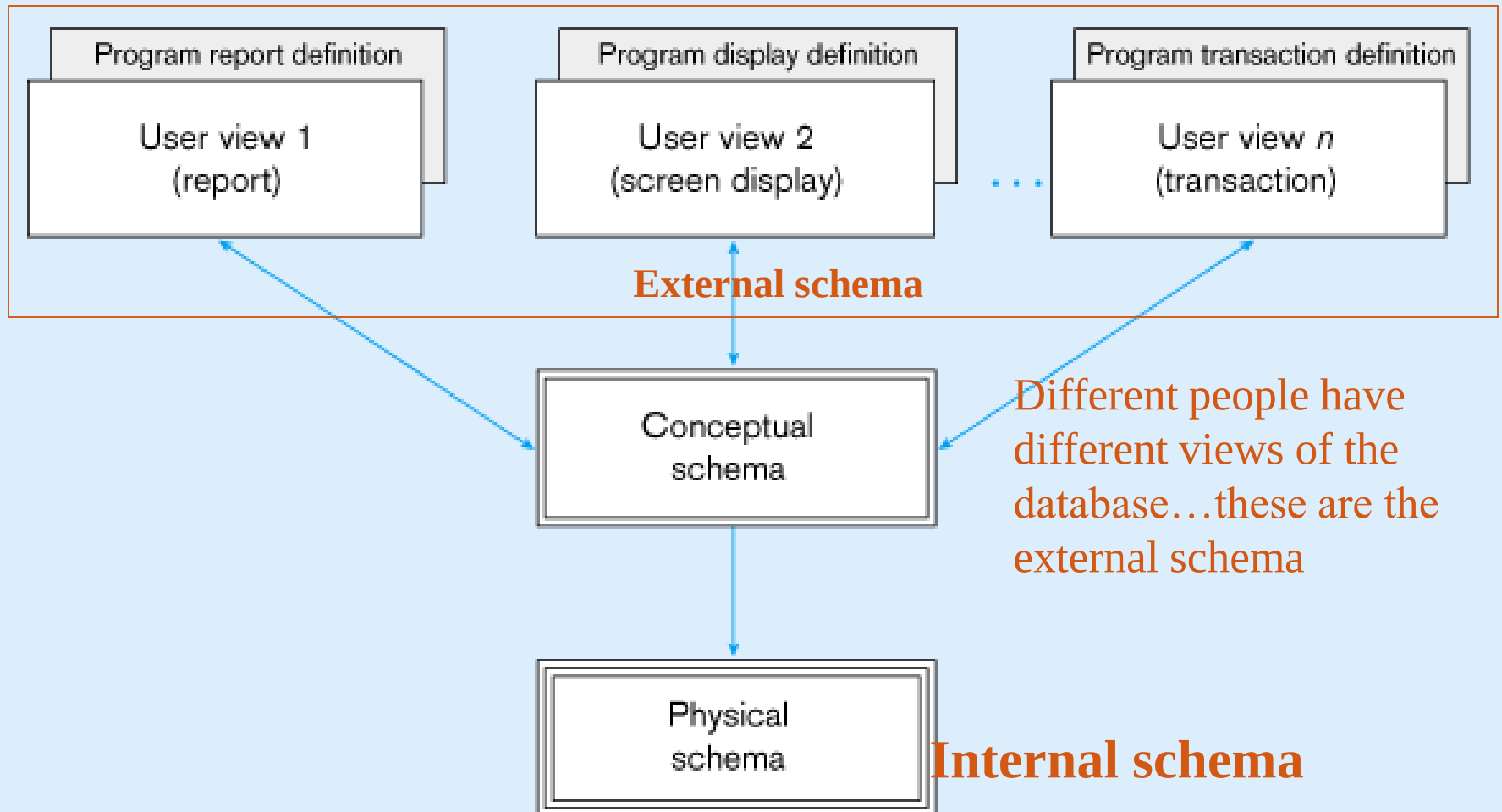
Systems programmers, network administrators, testers, technical writers

Gantt Chart



Three-schema database architecture

One organizational database description



THREE-SCHEMA ARCHITECTURE

- Goal is to separate the user application and the physical database:
 - **Internal schema**
 - **Conceptual schema**
 - **External schemas**

INTERNAL SCHEMA

- **Internal schema** describes physical storage structures and access paths
- Typically uses a *physical* data model and describes the complete detail of the data storage

CONCEPTUAL SCHEMA

Conceptual schema describes the structure and constraints for the *whole* database for a community of users.

Uses a *conceptual* data model which hides the details of physical storage structures and concentrates on describing

- Entities,
- data types,
- relationships,
- user operations,
- and constraints i.e. representational model

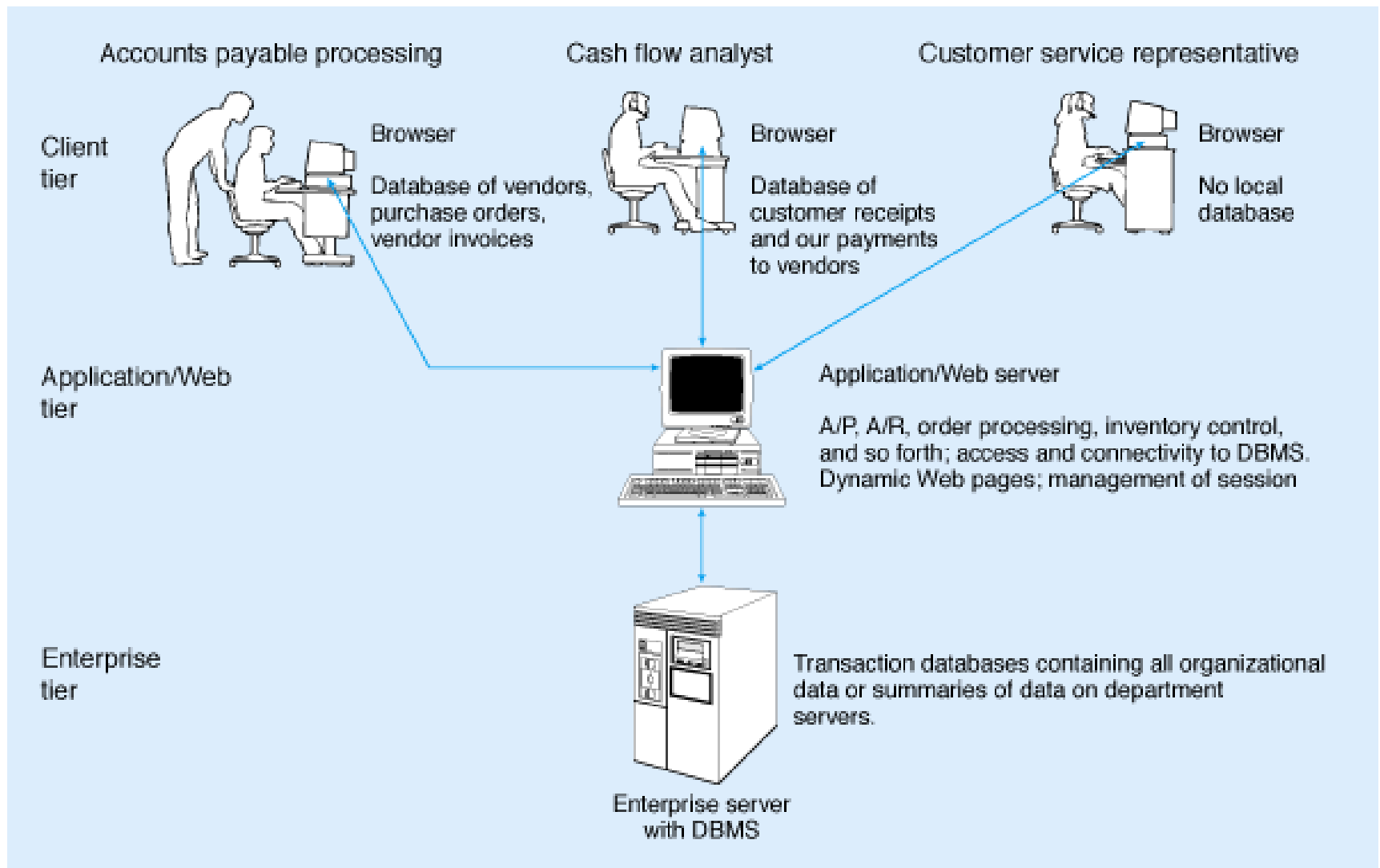
EXTERNAL SCHEMA

External schemas describes various user views

Usually uses the same data model as the conceptual level

Describes the part of the database that a particular user group is interested in and hides the rest of the database from that user group

Three-tiered client/server database architecture





Thanks